

Research Methodology in CS (CSEG3060_3)

Lecture 8: Regulatory Frameworks and Case Studies: Institutional Ethics Committees, plagiarism policies (UPES/academic bodies), and ethical dilemmas in research

Unit 3: Ethical Conduct and Literature Studies • Mapped Course Outcome: CO3

Lecture 8: Regulatory Frameworks and Case Studies

Course: Research Methodology in Computer Science (CSEG3060_3) **Program:** B.Tech (Computer Science) **Semester:** V **Unit:** 3 – Ethical Conduct and Literature Studies **Course Outcome:** CO3 – Demonstrate ethical conduct in research with adherence to academic integrity standards.

Estimated Reading Time: 60 minutes

1. Introduction

While Lecture 7 examined the philosophical underpinnings of research ethics, Lecture 8 transitions from theory to the operational and regulatory mechanisms that translate ethical principles into enforceable practice. In computer science research, where algorithms increasingly mediate decisions affecting human welfare, finance, healthcare, and civil liberties, the gap between ethical aspiration and regulatory enforcement is critically important. This lecture dissects the formal architecture of ethical oversight—Institutional Ethics Committees (IECs), university-level plagiarism policies, and national regulatory frameworks—and examines how these bodies adjudicate real-world dilemmas.

The discussion proceeds in four movements: (i) a survey of national and international regulatory frameworks governing research conduct; (ii) the structure, function, and decision-making processes of Institutional Ethics Committees; (iii) a detailed examination of plagiarism policies with specific reference to UPES and the University Grants Commission (UGC); and (iv) case studies that illustrate ethical dilemmas encountered by computer scientists.

2. Learning Objectives

Upon completion of this lecture, students will be able to:

1. **Identify** the principal national and international regulatory bodies governing research ethics in India.

- 2. **Describe** the composition, mandate, and review procedure of Institutional Ethics Committees.
- 3. **Apply** the UGC and UPES plagiarism regulations to concrete academic situations.
- 4. **Analyze** real-world case studies to detect ethical violations and recommend corrective action.
- 5. **Evaluate** emerging ethical dilemmas in computer science research, including those arising from artificial intelligence, big data, and generative systems.
- 6. **Formulate** a defensible position on contested ethical issues using established regulatory language.

3. Key Concepts at a Glance

Concept	Brief Definition
Institutional Ethics Committee (IEC)	A multidisciplinary body constituted to review, approve, and monitor research involving human participants or sensitive data.
Institutional Review Board (IRB)	The US equivalent of an IEC; often used interchangeably in international literature.
Regulatory Framework	A structured system of laws, guidelines, codes, and institutional policies governing research conduct.
Plagiarism	The appropriation of another's ideas, processes, results, or words without giving appropriate credit.
Similarity Index	A quantitative measure produced by plagiarism-detection software (e.g., Turnitin, iThenticate) expressed as a percentage of matched text.
Informed Consent	A process by which a participant voluntarily confirms his or her willingness to participate in research after being informed of all relevant aspects.
Research Misconduct	Fabrication, falsification, or plagiarism in proposing, performing, or reviewing research, or in reporting research results (US ORI definition).

4. Regulatory Frameworks Governing Research

4.1 International Frameworks

Several foundational documents shape global research ethics, including those in computer science.

4.1.1 The Nuremberg Code (1947)

Originating from the trials of Nazi physicians, the Nuremberg Code established ten principles for ethical human-subjects research, the foremost being **voluntary informed consent**. Although originally written for biomedical research, the principle of consent undergirds modern data-protection regulations, including those applied to user studies in computer science.

4.1.2 The Declaration of Helsinki (1964, last revised 2013)

Issued by the World Medical Association, this declaration extends the Nuremberg principles and distinguishes therapeutic from non-therapeutic research. It is the most widely adopted international standard for research involving human participants and is referenced by Indian institutional policies.

4.1.3 The Belmont Report (1979)

The US National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research articulated three principles—**Respect for Persons, Beneficence, and Justice**—that now inform most IEC review criteria.

4.1.4 ACM Code of Ethics (2018)

The Association for Computing Machinery's code outlines ethical responsibilities for computing professionals, including:

- Contribute to society and human well-being.
- Avoid harm.
- Be honest and trustworthy.
- Be fair and take action not to discriminate.
- Respect privacy.
- Honor confidentiality.

4.1.5 IEEE Code of Ethics

The IEEE Code commits members to "treat all persons fairly and with respect, to not engage in harassment or discrimination, and to avoid injuring others," and explicitly requires responsible disclosure of factors that might endanger the public.

4.2 Indian Regulatory Frameworks

India has developed a layered regulatory structure for research ethics:

4.2.1 ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants (2017, revised 2024)

The Indian Council of Medical Research (ICMR) guidelines are the most comprehensive Indian standard. The 2024 revision explicitly addresses **artificial intelligence in healthcare, big data analytics, and digital health technologies**. Section 12 of the guidelines introduces specific provisions for computational research, including requirements for algorithmic transparency, dataset representativeness, and bias mitigation.

4.2.2 University Grants Commission (UGC) Regulations

The UGC has issued two pivotal regulations:

- **UGC (Promotion of Academic Integrity and Prevention of Plagiarism) Regulations, 2018** – defines plagiarism levels and penalties in higher-education institutions.
- **UGC Guidelines on Academic Integrity and Plagiarism, 2024 (revised draft)** – updates similarity thresholds and incorporates provisions for generative AI content.

4.2.3 Central Drugs Standard Control Organization (CDSCO)

Oversees clinical trials and software-as-a-medical-device (SaMD), increasingly relevant for AI/ML-based diagnostic tools developed in computer science departments.

4.2.4 Ministry of Electronics and Information Technology (MeitY)

Issues policy frameworks on emerging technologies, including the National Strategy for Artificial Intelligence (2018) and the **Digital Personal Data Protection Act, 2023 (DPDP Act)**.

4.2.5 Information Technology Act, 2000 (and 2008 amendments)

Provides the legal foundation for prosecuting unauthorized data access, digital fraud, and cybercrime in India.

5. Institutional Ethics Committees (IECs)

5.1 Definition and Purpose

An Institutional Ethics Committee (IEC), sometimes called an Institutional Review Board (IRB) or Research Ethics Board (REB), is a formally constituted body that reviews research proposals to ensure they conform to ethical standards. The primary purposes are:

1. **Protection** of the rights and welfare of human participants.
2. **Assurance** that risks are minimized and reasonable in relation to anticipated outcomes.
3. **Verification** that informed consent procedures are adequate.
4. **Oversight** of ongoing research through continuing review.

5.2 Composition of an IEC

Per ICMR 2017 guidelines, an IEC must be multidisciplinary and gender-balanced, with a minimum of five and a maximum of twelve members. Typical composition includes:

- Chairperson (external to the institution, not affiliated with the organization)
- 1–2 basic medical scientists
- 1–2 clinicians
- One legal expert or retired judge
- One social scientist or philosopher
- One layperson representing community interests
- Member Secretary (who conducts the day-to-day administration)

For computer science research, an IEC should additionally include a **data protection officer** or **information security expert** when proposals involve large-scale personal data processing.

5.3 Levels of Review

IECs typically use a tiered review model:

Review Level	Trigger	Outcome
Exempt Review	Minimal risk; secondary use of de-identified data; survey of public officials	Notification of exemption
Expedited Review	Minimal risk but not exempt; minor modifications to approved protocols	Approval by Member Secretary and one member
Full Board Review	Greater than minimal risk; vulnerable populations; novel AI/ML systems involving personal data	Decision at convened meeting

5.4 The Review Process

The standard sequence is:

1. **Submission** – The principal investigator submits a protocol including the research proposal, consent forms, data management plan, recruitment materials, and risk-benefit analysis.
2. **Administrative Screening** – The Member Secretary checks for completeness.
3. **Primary Reviewer Assignment** – Two or more members conduct detailed reviews.
4. **Board Discussion** – At a convened quorum meeting, the investigator may be invited for questions.
5. **Decision** – Approval, modifications required, deferral, or disapproval.
6. **Notification** – A formal letter communicates the decision within a defined timeframe (typically 4–6 weeks).
7. **Continuing Review** – Annual progress reports; any amendment requires resubmission.

5.5 IEC Mandate at UPES

UPES University, Dehradun, maintains an **Institutional Ethics Committee (UPES-IEC)** that reviews research proposals involving:

- Human participants (surveys, interviews, behavioral experiments, usability studies).
- Personally identifiable information collected from data subjects.
- Animal subjects (less common in CS but required for HCI studies involving interaction with living systems in some labs).
- Environmental samples of biological origin.
- Dual-use research of concern (DURC).

Submissions to the UPES-IEC are routed through the online **Research Management System (RMS)** portal, and the committee meets quarterly with provision for emergency review of time-sensitive studies.

6. Plagiarism Policies

6.1 Defining Plagiarism

The UGC Regulations (2018) define plagiarism as:

"The practice of taking someone else's work or ideas and passing them off as one's own."

The definition encompasses:

- Verbatim copying of text without attribution.
- Close paraphrasing without acknowledgment.
- Uncited use of methodologies, datasets, or algorithms.
- Self-plagiarism (recycling one's own previously published work without disclosure).
- Collusion (coordinated submission of identical work).

6.2 Similarity Index Thresholds (UGC 2018)

Similarity Index	Category	Penalty (minimum)
Up to 10% (excluding quotes, bibliography, small matches)	Minor	Warning; no penalty if properly cited
10–40%	Moderate	Denial of degree for students; withdrawal of paper for faculty
Above 40%	Major	Expulsion; blacklisting of authors

6.3 The 2024 UGC Draft Revision

The 2024 draft proposes refinements, including:

- **Exclusion of generative AI output** from similarity detection.
- **Mandatory declaration** of AI-assisted content generation.
- **Stricter thresholds for dissertations** (5% for PhD theses without citation).
- **Specific treatment of code plagiarism** using tools such as MOSS (Measure of Software Similarity) or JPlag.

6.4 UPES Plagiarism Policy

UPES University adheres to the UGC framework with the following operational specifics:

1. **Software in Use:** UPES licenses **Turnitin** for textual submissions and **Ouriginal** for cross-checking. Code submissions may be checked using **MOSS** at the discretion of the supervising faculty.
2. **Acceptance Thresholds for Student Submissions:**
 - M.Tech/PhD dissertations: similarity index $\leq 10\%$ (excluding bibliography and quoted material).
 - B.Tech project reports: similarity index $\leq 20\%$.
 - Conference/journal publications from UPES: similarity index $\leq 15\%$ (excluding self-citation).

3. Submission Workflow:

- Step 1: Submit manuscript via the university LMS integrated with Turnitin.
- Step 2: Receive similarity report within 24 hours.
- Step 3: Supervisor reviews marked regions.
- Step 4: Revisions submitted if similarity exceeds threshold.
- Step 5: Final submission to the Departmental Research Committee (DRC).

4. Consequences of Non-Compliance:

- First offense: written warning and mandatory resubmission.
- Second offense: project grade of "F" and a six-month debarment from re-registration.
- Third offense or major plagiarism (>40%): expulsion and revocation of degree if discovered post-graduation.

5. Generative AI Policy: As of the 2024–2025 academic year, UPES permits the use of generative AI for grammar and code-suggestion purposes but requires explicit declaration in the Acknowledgement section. Submission of AI-generated text as the student's own original writing constitutes academic misconduct.

6.5 Other Academic Bodies and Their Plagiarism Norms

Body	Tool/Policy	Notable Provisions
AICTE	Recommends UGC framework	Mandatory for funded projects
IEEE	Self-declaration of originality	Use of AI tools must be disclosed
ACM	Author rights and ethics policy	Prohibits concurrent duplicate submission
Springer/Elsevier	iThenticate screening	Acceptability thresholds vary by journal
NAAC/NBA	Indirectly through institutional audits	Expects <15% across all UG/PG outputs

7. Ethical Dilemmas in Computer Science Research

While biomedical research dominates the historical literature on ethical dilemmas, computer science has generated a distinct set of contested issues.

7.1 Informed Consent in Online Studies

Unlike laboratory settings, online experiments are difficult to monitor. Key challenges:

- Participants may not fully read consent forms.
- Deception is sometimes necessary to study genuine behavior (e.g., studying susceptibility to phishing).
- Withdrawal of data after collection is technically difficult once results are anonymized and aggregated.

7.2 Data Privacy and Anonymization

Modern datasets from social media, mobile phones, and IoT devices often contain re-identifiable information. Simple anonymization is rarely sufficient; techniques such as differential privacy, k-anonymity, and federated learning have emerged to address this.

7.3 Algorithmic Bias and Fairness

Algorithms trained on biased data perpetuate and amplify social inequities. Examples include:

- Hiring algorithms that discriminate against women (e.g., the Amazon recruiting tool case, 2018).
- Facial recognition systems with higher error rates for women and people of color.
- Predictive policing algorithms that disproportionately target minority neighborhoods.

7.4 Dual-Use Research of Concern (DURC)

Research with legitimate civilian applications may also be weaponizable. Examples: large language models that generate disinformation, vulnerability discovery tools that facilitate cyberattacks, and AI systems that automate surveillance.

7.5 Authorship and Credit Disputes

The order of authors on a paper carries professional weight. Disputes arise when:

- A contributor is omitted despite substantive contribution (guest authorship, gift authorship).
- A senior researcher demands inclusion without contribution (honorary authorship).
- Two contributors claim priority for the same idea.

7.6 Whistleblowing and Disclosure

Researchers who discover unethical conduct by colleagues face a dilemma: loyalty to peers versus responsibility to the public. Many institutions, including UPES, have **whistleblower protection policies** aligned with the Public Interest Disclosure and Protection of Informers Act, 2010.

7.7 Reproducibility and Data Fabrication

The replication crisis has affected machine learning profoundly. Several high-profile retractions have occurred due to unverifiable results. Researchers sometimes face pressure to publish positive results, leading to selective reporting, p-hacking, or outright fabrication.

8. Case Studies

Case Study 1: The Facebook Emotional Contagion Study (2014)

Facts: Researchers at Cornell University, in collaboration with Facebook, manipulated the news feeds of approximately 689,000 users to study emotional contagion. Users were not asked for consent because the study was conducted under the platform's terms of service.

Ethical Issues:

- Lack of informed consent.
- Potential psychological harm.
- Insufficient IRB oversight (an external contractor was used).

Outcome: Public outcry led to regulatory scrutiny. The US Department of Health and Human Services clarified that certain social-media research does qualify as human-subjects research requiring IRB review. The lead researcher, Dr. Adam Kramer, issued a public apology.

Lesson: Even when research uses existing data, manipulation of user environments triggers ethical obligations. CS researchers must engage IECs for any study that alters user experience.

Case Study 2: The Cambridge Analytica Scandal (2018)

Facts: Personal data of approximately 87 million Facebook users was harvested through a personality-prediction app and used for political micro-targeting without consent.

Ethical Issues:

- Violation of informed consent.
- Misuse of data beyond stated purposes.
- Privacy violations on a mass scale.

Outcome: Investigations in the US, UK, and EU; congressional hearings; \$5 billion FTC fine on Facebook; the catalyst for the EU's enforcement of the General Data Protection Regulation (GDPR) and similar discussions leading to India's DPDP Act, 2023.

Lesson: Researchers must verify the chain of consent for any secondary data, particularly when data crosses jurisdictional lines.

Case Study 3: Theranos (2003–2018)

Facts: Theranos claimed to have developed a blood-testing technology requiring only a finger-prick sample. Internal whistleblowers alleged that the technology was non-functional and that test results were being run on conventional machines or fabricated.

Ethical Issues:

- Fabrication of data presented to investors and regulators.
- Endangering patient health through inaccurate diagnostics.
- Retaliation against whistleblowers.

Outcome: Criminal prosecution of the founders; SEC and FDA investigations; dissolution of the company.

Lesson: Data fabrication in computer science research (e.g., reporting model accuracy not supported by experiments) is research misconduct with severe consequences.

Case Study 4: HeLa Cells and the Henrietta Lacks Case

Facts: In 1951, Henrietta Lacks's cancer cells were taken without her knowledge or consent and became the basis for decades of biomedical breakthroughs. The HeLa genome was published in 2013 without consulting the Lacks family.

Ethical Issues:

- Original lack of informed consent.
- Commercial benefits accrued to companies, not the family.
- Privacy implications of genomic data publication.

Outcome: The NIH established a HeLa Genome Data Access Working Group including Lacks family members to govern future access. The case inspired data-access frameworks in computational biology.

Lesson: Biological and biomedical data derived from identifiable sources retain ethical obligations even decades later.

Case Study 5: AI Chatbot "Tay" by Microsoft (2016)

Facts: Microsoft released a Twitter chatbot that learned from interactions with users. Within 16 hours, coordinated adversarial input caused the bot to produce racist, misogynistic, and otherwise offensive tweets, leading to its shutdown.

Ethical Issues:

- Failure to anticipate adversarial manipulation.
- Inadequate pre-deployment testing.
- Lack of safety guardrails.

Outcome: Microsoft apologized and discontinued the product.

Lesson: Researchers deploying interactive AI systems must implement safeguards, conduct red-teaming exercises, and engage IEC review when human participants are involved in adversarial testing.

Case Study 6: Indian Domestic Plagiarism Incident at a Premier Institution (2016)

Facts: A doctoral scholar at an Indian IIT was found to have plagiarized approximately 60% of his thesis. The plagiarism was detected through Turnitin and confirmed by manual review.

Outcome: Per UGC Regulations, the scholar's registration was cancelled, and the supervising committee faced scrutiny for inadequate oversight.

Lesson: Supervisors bear responsibility for ensuring the originality of student work; institutions must conduct periodic checks throughout the research lifecycle, not only at submission.

9. Practical Example: Applying the UPES Plagiarism Policy

Consider a B.Tech student, Aditi, who submits her final-year project report to the UPES LMS for Turnitin screening. The report yields a 22% similarity index, with 14% attributable to a methodology section that closely paraphrases a published survey paper without quotation marks.

Step 1: Review. Aditi's supervisor notes the flagged regions. **Step 2: Threshold Check.** For B.Tech reports, the threshold is 20%. Aditi's 22% exceeds this. **Step 3: Remedial Action.** Aditi rewrites the methodology section, citing the original survey and using quotation marks for technical definitions. **Step 4: Resubmission.** The new similarity report shows 9%. **Step 5: Clearance.** The Departmental Research Committee accepts the revised report.

If Aditi had not revised the submission, her project would have received an "F" grade, and her supervisor would be required to report the incident to the Departmental Disciplinary Committee.

10. Summary

This lecture has examined the regulatory and institutional apparatus that operationalizes research ethics in computer science. The key takeaways are:

- Research ethics is enforced through a **multi-tiered regulatory architecture** spanning international codes (Nuremberg, Helsinki, Belmont), professional society codes (ACM, IEEE), national frameworks (ICMR, UGC, DPDP Act), and institutional bodies (IECs, IRBs).
 - **Institutional Ethics Committees** are the operational pivot of ethical review. Their composition, procedures, and authority are governed by ICMR and equivalent national standards. UPES operates the UPES-IEC with specific provisions for computer science research.
 - **Plagiarism policies** at UPES and under UGC guidelines use similarity-index thresholds and tiered consequences. The 2024 UGC draft revision introduces explicit treatment of generative AI.
 - **Ethical dilemmas in computer science** include informed consent in online studies, algorithmic bias, dual-use concerns, authorship disputes, and reproducibility crises.
 - **Case studies** illustrate the real-world consequences of ethical violations and underscore the necessity of institutional review.
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11. Exam Preparation Tips

1. **Memorize the specific thresholds.** UGC and UPES thresholds are frequently asked. Know the exact percentages and corresponding penalty levels.
2. **Differentiate review levels.** Understand when full-board review versus expedited review versus exemption applies.
3. **Be familiar with named documents.** Nuremberg Code, Declaration of Helsinki, Belmont Report, ICMR 2017, UGC 2018, DPDP Act 2023.
4. **Practice case analysis.** Examiners often present a hypothetical and ask students to identify the ethical issue and recommend a remedy.
5. **Know the composition of an IEC.** Be ready to explain why multidisciplinary composition matters.
6. **Link each dilemma to a specific code.** For instance, the Tay chatbot case implicates the ACM principle to "avoid harm."

7. **Stay current.** Emerging topics like generative AI and DPDP Act enforcement are likely to appear in current-affairs sections.
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12. Practice Questions

Short Answer Questions (2–3 marks each)

1. Define "Institutional Ethics Committee" and list three of its mandatory members as per ICMR 2017 guidelines.
2. State the UGC similarity-index threshold for "minor" plagiarism and the corresponding minimum penalty.
3. Differentiate between the Nuremberg Code and the Declaration of Helsinki.
4. What is dual-use research of concern (DURC)? Provide one example relevant to computer science.

Long Answer Questions (10 marks each)

1. Describe the composition, mandate, and review procedure of an Institutional Ethics Committee. How does this apply at UPES for computer science research?
2. Discuss the UGC Plagiarism Regulations (2018) and the proposed 2024 revisions. How does UPES operationalize these in its academic workflow?
3. Analyze the Facebook Emotional Contagion Study (2014) using the Belmont Report framework. What ethical principles were violated, and how could an IEC review have altered the study design?
4. Compare and contrast the ACM Code of Ethics (2018) with the IEEE Code of Ethics. To what extent are they enforceable in practice?
5. Examine the Cambridge Analytica case in the context of the Digital Personal Data Protection Act, 2023. What liabilities would researchers face if a similar incident occurred today at an Indian institution?

Multiple-Choice Practice

1. The Belmont Report articulates which three principles? (a) Honesty, Justice, Beneficence
(b) Respect for Persons, Beneficence, Justice
(c) Confidentiality, Autonomy, Non-maleficence
(d) Integrity, Transparency, Accountability
2. According to UGC 2018, similarity index above what percentage is classified as "major" plagiarism? (a) 20%

- (b) 30%
 - (c) 40%
 - (d) 60%
3. Which of the following is NOT typically a member of an IEC? (a) Legal expert
(b) Layperson from the community
(c) Marketing executive of the sponsor
(d) Clinician or medical scientist
4. The HeLa cells case primarily deals with which ethical issue? (a) Data fabrication
(b) Informed consent and data ownership
(c) Dual-use research
(d) Algorithmic bias
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13. References

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End of Lecture 8 Notes